

Remark 1.11. This is an adaptation of an established certification result, not a new theorem — the correspondence between hull thinness and $\bar{p}$ is heuristic (they are related but not identical objects: one lives in representation space, the other in input space). We include it because it is the cleanest existing bridge between “geometric fingerprint of vulnerability” and a certifiable robustness guarantee, and it suggests a concrete follow-up experiment: regress empirical certified radius against I_ℓ across the architecture sweep already run for the CIFAR-10/SVHN figures.

Remark 1.12 (Limitation). I_ℓ is a second-moment surrogate for hull geometry. It cannot detect a success region that is genuinely multi-modal (e.g., two disjoint thin clusters in orthogonal directions) but has zero net correlation — such a region would register the same ER as a single isotropic-in-2D cluster. This is the same blind spot every covariance-based metric inherits, including the PCA Score used in Contribution II, and is worth stating explicitly rather than leaving implicit.

2 Contribution II Theory: A Spectral Theory of the Structural Bottleneck Score

2.1 Setup

Let $\Phi : \mathbb{R}^{d_{in}} \rightarrow \mathbb{R}^{d_{out}}$ be a layer or branch map (a residual block, a neural-plate branch, etc.) acting on data distribution $\mathcal{D}$. Let X_{in} be a sample of pre-activation representations and $X_{out} = \Phi(X_{in})$, with covariances $\Sigma_{in}, \Sigma_{out}$ and eigenvalues $\lambda_1^{in} \geq \lambda_2^{in} \geq \dots, \lambda_1^{out} \geq \lambda_2^{out} \geq \dots$.

Definition 2.1 (PCA Score). *For covariance Σ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_d$,*

$$\text{PCA}(\Sigma) = \frac{\lambda_1}{\sum_{i \geq 2} \lambda_i} \in [0, \infty).$$

PCA(Σ) measures dominance of the leading direction relative to the residual spectrum, and diverges as the spectrum degenerates toward rank 1.

Remark 2.2. ER (Definition 1.2) and PCA are both spectral-concentration statistics but are not interchangeable: ER is a global entropy functional sensitive to the *shape* of the entire tail, while PCA is a local ratio sensitive only to how much the top direction dominates everything else combined. The Structural Bottleneck Score below deliberately uses one of each, giving it two independent lenses on the spectrum rather than two correlated measurements of the same thing.

Definition 2.3 (Rank Contraction and Concentration Gain).

$$\rho(\Phi) := 1 - \frac{\text{ER}(\Sigma_{out})}{\text{ER}(\Sigma_{in})}, \quad \kappa(\Phi) := \text{PCA}(\Sigma_{out}) - \text{PCA}(\Sigma_{in}).$$

Definition 2.4 (Structural Bottleneck Score).

$$S(\Phi) := \rho(\Phi) \cdot \kappa(\Phi).$$

Remark 2.5 (Sign convention). The source material writes this informally as $(1 - \text{rank}(\text{in})/\text{rank}(\text{out}))$, which is sign-ambiguous: if the intended “rank” is effective rank and Φ contracts it ($\text{ER}_{out} < \text{ER}_{in}$), the literal in/out ratio exceeds 1 and the bracket goes *negative* — contradicting the stated property “higher S = stronger bottleneck.” We adopt the orientation above $(1 - \text{ER}_{out}/\text{ER}_{in})$ precisely because it is the one consistent with both stated properties (“ $S = 0$ at identity,” “higher $S \Rightarrow$ stronger bottleneck”); this should be checked against the implementation before this section goes to print, since the two conventions differ by more than a sign flip in general.

2.2 Basic Properties

Proposition 2.6 (Identity Vanishing). *If $\Phi = \text{id}$, then $\Sigma_{\text{out}} = \Sigma_{\text{in}}$, hence $\text{ER}_{\text{out}} = \text{ER}_{\text{in}}$, so $\rho(\Phi) = 0$ and $S(\Phi) = 0$. More generally, $S(\Phi) = 0$ for any Φ that preserves the eigenvalue multiset of the covariance (e.g. any map that is, up to an orthogonal transform, an isometry on the data manifold), regardless of $\kappa(\Phi)$.*

Proof. Immediate from Definition 2.3. □

Proposition 2.7 (Boundedness and a Numerical Caveat). *Since $\text{ER}(\Sigma) \in [1, d]$ for any Σ , $\rho(\Phi) \in [1 - d_{\text{out}}, 1)$. However $\kappa(\Phi)$ is unbounded above: $\text{PCA}(\Sigma) \rightarrow \infty$ as the spectral tail $\sum_{i \geq 2} \lambda_i \rightarrow 0$, i.e. exactly in the near-total-collapse regime the score is designed to reward most. Consequently $S(\Phi)$ is bounded only away from this degenerate limit.*

Remark 2.8 (Implementation note). This is not a defect of the theory but a numerical-stability issue worth flagging: any practical implementation should regularize the denominator, e.g. $\text{PCA}_\eta(\Sigma) = \lambda_1 / (\eta + \sum_{i \geq 2} \lambda_i)$ for small $\eta > 0$, or cap S during training-time monitoring. Otherwise a branch that collapses to (near-)rank-1 can produce an arbitrarily large S that swamps the growth-decision comparison against threshold τ .

2.3 A Duality Theorem: Why the Product Form

Theorem 2.9 (Rank-Concentration Duality).

$$S(\Phi) > 0 \iff \text{sign}(\rho(\Phi)) = \text{sign}(\kappa(\Phi)) \neq 0.$$

Equivalently: Φ registers as a genuine structural bottleneck exactly when rank contraction and spectral concentration co-occur ($\rho, \kappa > 0$), or when both simultaneously reverse ($\rho, \kappa < 0$; see Remark 2.10). A layer that contracts rank while dispersing variance ($\rho > 0, \kappa < 0$), or expands rank while concentrating variance ($\rho < 0, \kappa > 0$), contributes $S \leq 0$.

Proof. Immediate sign algebra on a product of two reals. □

Remark 2.10 (Why multiplicative, not additive). This gives a principled reason to prefer the product form $S = \rho \cdot \kappa$ over the more naive sum $\rho + \kappa$. An additive score can be strongly positive even when the two signals *disagree* — e.g. large rank contraction paired with a collapsing (not concentrating) spectrum tail would still push $\rho + \kappa$ positive if ρ dominates, falsely registering a bottleneck. The product requires the two pieces of evidence to agree in sign before contributing a positive score at all. This is the theoretical justification the multiplicative form on Slide 6 was missing.

Remark 2.11 (A genuine blind spot). The case $\rho < 0, \kappa < 0$ also yields $S > 0$: a branch that simultaneously *expands* effective rank and *loses* concentration — becoming more distributed and more polysemantic on both axes — registers the same positive score as a genuine bottleneck. This is a real gap in Definition 2.4 as it stands, not a modeling choice. Since the growth-decision rule only ever needs to spawn capacity when there is a contraction to compensate for, we recommend gating the criterion by $\mathbb{I}[\rho(\Phi) > 0]$ — i.e., trigger growth on $S(\Phi) \cdot \mathbb{I}[\rho(\Phi) > 0] > \tau$ rather than on $S(\Phi) > \tau$ alone. This is a one-line fix and is worth deciding on before it is stated as settled in the paper.

2.4 Relating the Growth Threshold to Capacity Sufficiency

Proposition 2.12 (Threshold as a Capacity Statement). *Fix $\tau \in (0, 1)$. Then*

$$\rho(\Phi) \geq \tau \iff \text{ER}(\Sigma_{in}) \geq \frac{\text{ER}(\Sigma_{out})}{1 - \tau}.$$

Proof. Direct algebra on Definition 2.3: $1 - \text{ER}_{out}/\text{ER}_{in} \geq \tau \iff \text{ER}_{out}/\text{ER}_{in} \leq 1 - \tau \iff \text{ER}_{in} \geq \text{ER}_{out}/(1 - \tau)$. $\square$

Remark 2.13. This is an equivalent restatement, not a deep result, but it gives the growth threshold $\tau = 0.1$ a concrete reading: triggering growth when $\rho \geq 0.1$ says the incoming representation carries at least $\sim 11\%$ more effective dimensionality than the branch’s current output retains — the branch is being asked to represent more independent directions of variation than its current capacity supports. Combined with Theorem 2.6, the *full* growth criterion $S > \tau$ requires this capacity shortfall to be accompanied by genuine concentration of what *is* retained.

Remark 2.14 (Relation to the information bottleneck; an open limitation). The two-term structure of S echoes Tishby’s information-bottleneck objective — trading compression of the input against sufficiency of the retained signal — but S is entirely unsupervised: it has no term analogous to $I(T; Y)$ and so cannot distinguish a compression that discards nuisance variance from one that discards task-relevant variance. A branch could score well on S while quietly destroying label-relevant information. This is worth stating as an open problem, since it is exactly the kind of question a reader familiar with the information-bottleneck literature will ask.

Note on How These Connect

Both frameworks are built from the same two spectral primitives (ER, PCA), which is worth saying explicitly in the paper: Contribution I applies them to the covariance of a *sampled adversarial hull*, and Contribution II applies them to the covariance of *ordinary forward activations* at input and output of a branch. Under the identification in Theorem 1.5, both scores are reading off the same underlying latent variable — the effective dimensionality of feature superposition — from two different geometric objects. That shared machinery is the actual content of the claim “these are two sides of the same coin,” and stating it this way gives the reader something falsifiable: if I_ℓ and S are truly measuring the same latent variable, they should be strongly correlated layer-by-layer across the architecture sweep already collected for the empirical figures — a direct, cheap check against existing data before the paper claims the connection.